Divisibility and Congruences

Number Systems: From $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 2 built $\mathbb{Z}$ out of pairs of natural numbers and proved it an ordered ring, but it said nothing about which integers divide which. *Sets and Functions*, Section 3.2 did, for the school integers: it proved division with remainder, showed that every integer from 2 on has a prime factor, and set the uniqueness of prime factorization aside as a fact it neither proved nor used. This chapter proves that arithmetic again for the constructed $\mathbb{Z}$, and goes further, using the ordered-ring axioms and two properties of the order of $\mathbb{Z}$: there is no integer strictly between 0 and 1 (Number Systems: From \(\nN\) to \(\nR\), Theorem 2.4.2 (c)), and a non-empty set of integers with a lower bound has a least element (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.4); the field $\mathbb{Q}$ enters chiefly for the integer part $\lfloor a/n \rfloor$ and for lowest terms. The second property gives division with remainder at once. Then one move does most of the work. In a set of integers closed under sums and under multiplication by any integer, the **least positive element** divides every element, and dividing by it with remainder is how one sees this. Applied to the combinations $ax + by$, it makes the greatest common divisor a combination of a and b (Bézout's theorem), and the Euclidean algorithm computes it. Multiplying that combination by a third integer gives Euclid's lemma, that a prime dividing a product divides one of the factors, and with it the uniqueness of prime factorization and the lowest terms of a rational number.

The second half returns to the clock of *Sets and Functions*, Section 5.2. There, and again for $\mathbb{Z}$ and $\mathbb{Q}$ in *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapters 2 and 3, each sum and product of classes was checked separately to be well defined. Section 1.3 makes the check once, for the quotient of any commutative ring by an ideal, and $\mathbb{Z}/n\mathbb{Z}$ becomes a commutative ring with n elements on which, for $n \geq 2$, no order fits. In it the classes that can be divided by are exactly the classes of the integers coprime to n. So $\mathbb{Z}/p\mathbb{Z}$ is a field with p elements for every prime p, and two congruences with coprime moduli have exactly one class of solutions modulo the product of the moduli, the Chinese remainder theorem. *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 5 builds $\mathbb{R}$ in the same shape, as classes of rational Cauchy sequences whose difference is null (the null sequences play the part of the ideal), though it checks its operations directly; Chapter 5 of this volume runs division with remainder, gcd and unique factorization again for polynomials over a field.

Two integers, one theory. *Sets and Functions*, Chapters 1 and 3 proved its facts about divisibility for the school integers, from school facts such as the closure of $\mathbb{Z}$ under $+$, $-$ and $\cdot$ (Sets and Functions, Proposition 3.1.2). This chapter keeps their definitions of divisibility, prime and congruence (Sets and Functions, Definition 1.1.2, Sets and Functions, Definition 1.1.3, Sets and Functions, Definition 5.2.5), read word for word on the ring of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 2, and proves the results again; no proof in it cites a result of *Sets and Functions* about integers. Each such result has a twin here: transitivity of divisibility (Sets and Functions, Proposition 1.1.1) and even or odd (Sets and Functions, Theorem 1.1.2), from its Chapter 1, are Number Systems: From \(\nN\) to \(\nR\), Proposition 2.5.1 (b) and Number Systems: From \(\nN\) to \(\nR\), Corollary 2.5.3 (a); the others are named in the section that proves them: division with remainder (Sets and Functions, Theorem 3.2.3) and the antisymmetry of divisibility on $\mathbb{N}$ (Sets and Functions, Proposition 5.1.1) in *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Section 2.5; a prime factor of every $n \geq 2$ (Sets and Functions, Theorem 3.2.5) and lowest terms (Sets and Functions, Lemma 3.2.4) in Section 1.2; and the arithmetic of congruence classes (Sets and Functions, Proposition 5.2.5) in Section 1.3. Nothing earlier was circular: the earlier chapters used these facts about the school integers as stated facts, and this chapter proves them for the constructed ones.

What you need. *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapters 1–3, chiefly Sections 2.3–2.5,

and *Sets and Functions*, Chapters 1–5, chiefly Sections 1.1, 3.2 and 5.2. From *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapters 2 and 3: the commutative ring $\mathbb{Z}$ and its algebra (Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1, Number Systems: From \(\nN\) to \(\nR\), Theorem 2.3.1, Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2), its order, its positive elements, the rules for inequalities and discreteness (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.1, Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.3, Number Systems: From \(\nN\) to \(\nR\), Theorem 2.4.2), the least element of a set of integers bounded below (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.4), the copy of $\mathbb{N}$ inside $\mathbb{Z}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.2.3), the lemma that a rule on classes is a function when equivalent inputs give equivalent outputs (Number Systems: From \(\nN\) to \(\nR\), Theorem 2.2.1), the two-element field that cannot be ordered (Number Systems: From \(\nN\) to \(\nR\), Example (A field with two elements that cannot be ordered)), and the field $\mathbb{Q}$ with $\mathbb{Z}$ inside it, its order, reciprocals and integer part (Number Systems: From \(\nN\) to \(\nR\), Definition 3.1.2, Number Systems: From \(\nN\) to \(\nR\), Theorem 3.1.3, Number Systems: From \(\nN\) to \(\nR\), Proposition 3.1.4, Number Systems: From \(\nN\) to \(\nR\), Theorem 3.2.1, Number Systems: From \(\nN\) to \(\nR\), Proposition 3.2.2, Number Systems: From \(\nN\) to \(\nR\), Proposition 3.2.5); and from its Section 2.5, division with remainder, the rules for divisibility and parity (Number Systems: From \(\nN\) to \(\nR\), Theorem 2.5.2, Number Systems: From \(\nN\) to \(\nR\), Proposition 2.5.1, Number Systems: From \(\nN\) to \(\nR\), Corollary 2.5.3). From its Chapter 1, the citing rule of its Section 1.4 (Number Systems: From \(\nN\) to \(\nR\), Corollary 1.4.3), by which induction (Logic and Proof, Axiom 4.2.1) and the recursion theorem (Sets and Functions, Theorem 3.3.1, Number Systems: From \(\nN\) to \(\nR\), Theorem 1.2.1) apply to $\mathbb{N}$; and powers and parity on $\mathbb{N}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 1.3.4), used in review exercise R2. From *Sets and Functions*, Section 1.1, the definitions of divisibility and prime (Sets and Functions, Definition 1.1.2, Sets and Functions, Definition 1.1.3); from its Section 3.3, finite products (Sets and Functions, Definition 3.3.1), used in Section 1.2 as a pattern for products of lists; from its Section 5.2, equivalence relations, classes and quotient sets (Sets and Functions, Definition 5.2.1, Sets and Functions, Definition 5.2.4, Sets and Functions, Theorem 5.2.1) and congruence (Sets and Functions, Definition 5.2.5); from its Section 5.3, suprema and the divisibility order (Sets and Functions, Definition 5.3.5, Sets and Functions, Example (The supremum under divisibility is the least common multiple)), in a remark of Section 1.1 and in Section 1.2's B2. From *Sets and Functions*, Chapters 1 and 2: products, equality of functions and bijections (Sets and Functions, Definition 1.4.2, Sets and Functions, Proposition 2.1.1, Sets and Functions, Definition 2.2.3); and from its Section 4.1, finite sets and equinumerous sets (Sets and Functions, Definition 4.1.2, Sets and Functions, Proposition 4.1.1), to count $\mathbb{Z}/n\mathbb{Z}$, with more on finite sets (Sets and Functions, Proposition 4.1.6, Sets and Functions, Corollary 4.1.4, Sets and Functions, Theorem 4.1.9, Sets and Functions, Exercise (C2: Counting products and functions), Sets and Functions, Corollary 4.1.3) in Section 1.4's B3 and C2. From *Logic and Proof*, Chapter 8, the catalogue of what a proof may not do (Logic and Proof, Example (What is not allowed)), for Section 1.2's B1 and C2 and Section 1.4's C2. A few results are compared with and not used: the twins listed above; Euclid's argument in *Logic and Proof*, Section 8.1's C1 (Logic and Proof, Exercise (C1: Euclid without the contradiction)) in Section 1.2's C3; a rule on classes that is not a function (Sets and Functions, Example (A rule on classes that is not a function)) in Section 1.3's C1; lowest terms in *Sets and Functions*, Section 3.2's C2 (Sets and Functions, Exercise (C2: Lowest terms, used later)) in Section 1.2; and, in the review exercises, the countable product of *Sets and Functions*, Section 4.2 (Sets and Functions, Exercise (B2: Putting a proof in order), Sets and Functions, Theorem 4.2.6) in R2, the two proofs that $\sqrt{2}$ is not rational (Logic and Proof, Theorem 2.2.1, Number Systems: From \(\nN\) to \(\nR\), Theorem 3.3.1) in R3. Review exercise R1 also returns to the integers as classes of pairs (Number Systems: From \(\nN\) to \(\nR\), Definition 2.1.1).

Time. About 25 hours on the core track (all of A, one B and one C per section, the exit checks); about 30 hours for everything.

1.-1.1 Roadmap

- **The Greatest Common Divisor, Bézout, and the Euclidean Algorithm.** The combinations $a\mathbb{Z}+b\mathbb{Z}$ and the multiples $n\mathbb{Z}$, the greatest common divisor with $\gcd(0,0) \coloneqq 0$, Bézout's theorem (the least positive combination is the gcd, and every combination is a multiple of it), coprime integers, and the Euclidean algorithm as a recursion that keeps the set of combinations fixed, with back-substitution for the coefficients.
- **Euclid's Lemma and the Fundamental Theorem of Arithmetic.** Euclid's lemma from the Bézout identity, a prime dividing a product of primes equal to one of them, coprime divisors, the least divisor above 1 as a prime, existence and uniqueness of non-decreasing prime factorizations by comparing the largest primes, a monoid in which factorization is not unique, and lowest terms in $\mathbb{Q}$.
- **Ideals, Quotient Rings, and the Integers Modulo n.** Ideals as the subsets that a quotient needs, the quotient ring R/I of a commutative ring by an ideal, the failure for $\mathbb{Z} \subseteq \mathbb{Q}$, the ideals of $\mathbb{Z}$ as the sets $n\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ as a commutative ring with n elements that no order fits when $n \geq 2$.
- **Units, Prime Moduli, and the Chinese Remainder Theorem.** Units of a commutative ring, the units of $\mathbb{Z}/n\mathbb{Z}$ as the classes coprime to n with inverses from back-substitution, $\mathbb{Z}/n\mathbb{Z}$ a field exactly when n is prime, and the Chinese remainder theorem for two coprime moduli, with the failures when the moduli share a factor; the review exercises reach back to *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapters 1–3, *Logic and Proof*, Chapter 2, and *Sets and Functions*, Chapter 4.

1.-1.2 Named moves

- **Least positive element, then divide.** Take the least positive element d of a set of integers closed under integer combinations; divide any element of the set by d; the remainder lies in the set and is smaller than d, so it is 0. This proves Bézout's theorem in Section 1.1 and the description of the ideals of $\mathbb{Z}$ in Section 1.3, and Section 1.2's B2 and Section 1.3's C2 use it for common multiples and for sets closed under subtraction.
- **Multiply the Bézout identity.** From $1 = ax + by$, multiply by c: $c = acx + bcy$, and read off what divides c. It turns a statement about common divisors into an equation, and it gives Euclid's lemma in Section 1.2; in Section 1.4 the same identity, read modulo n, gives inverses, and read modulo m and modulo n at once, the Chinese remainder theorem.
- **Compute on representatives; the ideal absorbs the change.** An operation on classes $a + I$ is well defined because a change of representative is an element of I, and multiplying it by the other argument keeps it in I. This is *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 2's check one argument at a time, now made once for every commutative ring and every ideal.